

Oculoglandular Tularemia — Comprehensive Disease Characterization Report

Disease: Oculoglandular Tularemia **MONDO ID:** MONDO:0001665 **Category:** Infectious Disease (bacterial zoonosis) **Causative agent:** *Francisella tularensis* (NCBI Taxon: 263) **Evidence base:** Aggregated disease-level literature (case reports, case series, systematic reviews, and mechanistic model-organism/in-vitro studies). No individual EHR-level or omics data files were provided; this is a literature-synthesis report.

Summary

Oculoglandular tularemia is the conjunctival-entry form of tularemia, an acute zoonotic infection caused by the Gram-negative, facultatively intracellular bacterium *Francisella tularensis*. It is one of the six classic clinical forms of tularemia and accounts for approximately **1–3% of all tularemia cases**. Clinically it presents as unilateral granulomatous follicular conjunctivitis accompanied by painful ipsilateral preauricular, submandibular and cervical lymphadenopathy — the constellation known as **Parinaud oculoglandular syndrome (POGS)**. Among causes of POGS, oculoglandular tularemia ranks **second only to cat scratch disease** (*Bartonella henselae*), which is the single most important differential diagnosis. Infection follows conjunctival inoculation by contaminated fingers, aerosol splashes, or ocular trauma involving infected animal material.

The pathophysiology is that of an intracellular bacterial infection rather than a heritable disorder — there is **no host causal gene, inheritance pattern, or germline variant**. After conjunctival inoculation, *F. tularensis* is phagocytosed by macrophages, escapes the phagosome using a non-canonical Type VI secretion system (T6SS) encoded on the Francisella Pathogenicity Island (FPI; effectors PdpC/PdpD), replicates in the cytosol, and triggers the AIM2 inflammasome with caspase-1 activation, IL-1 β /IL-18 release, and pyroptotic cell death. This drives granulomatous conjunctival inflammation and regional suppurative lymphadenitis. *F.*

tularensis is one of the most infectious bacteria known, requiring as few as ~10–15 organisms to cause disease, which underlies its classification as a Tier 1 select agent and potential bioterrorism threat.

Diagnosis relies on **serology (microagglutination titer $\geq 1/160$ or four-fold rise)** and **PCR** (targets *tul4*, *ISFtu2*, *fopA*, 23-kDa), because culture is insensitive (~10% positive) and hazardous (requires BSL-3 containment). The disease is highly curable: contemporary isolates remain uniformly susceptible to aminoglycosides, fluoroquinolones, and tetracyclines, while β -lactams reliably fail. Case-fatality among appropriately treated patients is under 2%, and *F. tularensis* subsp. *holarctica* (the European/Asian type B) infections are essentially non-fatal. Suppurative lymph nodes occasionally require surgical drainage. Prevention is behavioral/environmental (personal protective equipment, avoiding sick/dead lagomorphs and rodents, tick-bite avoidance, hand and eye hygiene) plus antibiotic post-exposure prophylaxis; **no licensed human vaccine exists**.

Key Findings

Finding 1 — Definition and rank among causes of Parinaud oculoglandular syndrome

Oculoglandular tularemia is a rare clinical form of tularemia and the **second commonest cause of Parinaud oculoglandular syndrome (POGS)**. POGS is defined as unilateral granulomatous palpebral conjunctivitis with ipsilateral preauricular, submandibular, and cervical lymphadenopathy, frequently accompanied by fever and malaise. Reviews of POGS etiology consistently identify cat scratch disease (*Bartonella henselae*) as the most common cause, with oculoglandular tularemia second. Entry is via conjunctival inoculation — contaminated fingers after handling infected animals, aerosol splashes, or ocular trauma from infected animal material.

"The most common underlying pathology is cat scratch disease, followed by the oculoglandular form of tularemia." — [PMID: 38941282](#)

"Parinaud's oculoglandular syndrome should be considered in the differential diagnosis of a patient presenting with unilateral granulomatous conjunctivitis, painful preauricular, and submandibular lymphadenopathy combined with systemic symptoms of general malaise and fever. Tularemia is one etiology of Parinaud's oculoglandular syndrome." — PMID: 11228320

Finding 2 — Uniform antibiotic susceptibility of *F. tularensis*

Francisella tularensis isolates remain **uniformly susceptible to first-line tularemia antibiotics**. Antimicrobial susceptibility testing of 278 U.S. isolates (2009–2018) against 8 drugs (ciprofloxacin, levofloxacin, doxycycline, tetracycline, gentamicin, streptomycin, chloramphenicol, erythromycin) found **all isolates susceptible to all drugs tested**. German subsp. *holarctica* isolates (n=128) showed low MIC90 values: gentamicin 1 mg/L, streptomycin 4.0 mg/L, tetracycline 0.5 mg/L, doxycycline 1.5 mg/L, and ciprofloxacin 0.064 mg/L. One important exception: subsp. *holarctica* biovar II (genotype B.12) is intrinsically erythromycin-resistant. Streptomycin (intramuscular) is the historical drug of choice; ciprofloxacin was used successfully in a confirmed oculoglandular case after β -lactam failure.

"We tested the susceptibility of 278 *F. tularensis* isolates from the United States received during 2009-2018 to 8 antimicrobial drugs... All isolates were susceptible to all tested drugs." — PMID: 38294116

"empiric ciprofloxacin therapy was administered, and the patient recovered without sequelae" — PMID: 38941282

Finding 3 — Epidemiology, infectivity, and low treated fatality

Oculoglandular tularemia is an uncommon form (~1–3% of cases), and *F. tularensis* is extraordinarily infectious with low fatality when treated. A systematic review of 870 human tularemia cases (1993–2023, 35 countries) noted that the organism "requires as few as 10 organisms to cause disease," and reported case-fatality among treated patients of **0.7% (aminoglycosides), 0.9% (fluoroquinolones), and 1.2% (tetracyclines)**. In a Missouri series of 121 cases (2000–2007), the oculoglandular form comprised **3%** of cases (vs. ulceroglandular 37%, glandular 25%, pneumonic 12%, typhoidal 10%, oropharyngeal 2%); median incubation was 3 days (range 1–9), 65% of patients were male, median age was 37, and 69% of cases were attributed to tick bites, with systemic disease more common in older patients.

Clinical form	Frequency (Missouri, n=121)
Ulceroglandular	37%
Glandular	25%
Pneumonic	12%
Typhoidal	10%
Oculoglandular	3%
Oropharyngeal	2%

"requires as few as 10 organisms to cause disease, making this potential bioterrorism agent one of the most infectious bacterial pathogens known" — [PMID: 38294108](#)

"the fatality rate was 0.7%, 0.9%, and 1.2%, respectively" — [PMID: 38294108](#)

"Most patients presented with ulceroglandular (37%) and glandular (25%) forms of tularemia, followed by pneumonic (12%), typhoidal (10%), oculoglandular (3%), and oropharyngeal (2%) forms." — [PMID: 22911645](#)

Finding 4 — Pathogenesis causal chain

The pathogenesis proceeds from conjunctival inoculation to clinical disease through a well-characterized intracellular mechanism. *F. tularensis* is a facultative intracellular pathogen that survives within phagocytic cells "through phagosomal escape and replication in the cytosol, ultimately causing inflammasome activation and host cell death." Phagosomal escape requires a **non-canonical T6SS encoded on the 33-kb Francisella Pathogenicity Island (FPI)**; effectors PdpC and PdpD are required for phagosome rupture, and strains lacking them cannot escape the phagosome, activate the AIM2 inflammasome, or cause disease in mice. Cytosolic bacterial DNA activates the AIM2 inflammasome, leading to autocatalytic caspase-1 cleavage, IL-1 β and IL-18 secretion, and pyroptotic cell death. Host NLRP3 increases susceptibility to tularemia in the mouse. Histopathology of oculoglandular lesions shows granulomatous inflammation with necrosis.

"strains lacking pdpC and pdpD are unable to escape from phagosome, activate AIM2 inflammasome or cause disease in mice. This suggests that PdpC and PdpD are T6SS effectors involved in phagosome rupture." — [PMID: 28621333](#)

"Its life cycle is characterized by an ability to survive within phagocytic cells through phagosomal escape and replication in the cytosol, ultimately causing inflammasome activation and host cell death." — [PMID: 27830989](#)

"Escape of *E. tularensis* from the phagosome into the cytosol of the macrophage triggers the activation of the AIM2 inflammasome" — [PMID: 23115038](#)

Finding 5 — Diagnosis via serology and PCR; culture insensitive

Diagnosis relies on **serology (microagglutination)** and **PCR**; culture is insensitive (~10%) and biohazardous. A microagglutination titer $\geq 1/160$ in a single serum, or a ≥ 4 -fold rise/seroconversion between paired sera, is diagnostic — in one Turkish outbreak, 68% were positive on first serum and all initially negative patients seroconverted. PCR on lymph-node aspirate targeting *tul4*, *ISFtu2*, *fopA* and 23-kDa genes detects and subspecies the organism (91% PCR-positive on node aspirates in one series); a multitarget TaqMan assay (*ISFtu2*, 23-kDa, *tul4*) reached a 1-organism detection limit and outperformed culture. Culture is positive in only ~10% of cases and requires BSL-3 containment; MALDI-TOF and 16S rRNA sequencing identify isolates. Failure to respond to β -lactams is a key diagnostic clue. The chief differential is cat scratch disease (*Bartonella henselae*).

"in 17 (68%) of them microagglutination test yielded positive result ($\geq 1/160$) in their first serum samples. All of the 8 patients who had negative results in their first samples ($< 1/160$), revealed seroconversion in their second samples. In 10 (91%) of the 11 patients from whom lymph node aspirates were obtained, PCR performed with species specific (*tul4*) primers yielded positivity" — [PMID: 22090310](#)

"the TaqMan PCR assay was significantly more sensitive than culturing" — [PMID: 14662930](#)

"positive cultures are typically obtained in only 10% of tularemia cases" — [PMID: 37209668](#)

"The primary inoculation complex causing regional lymphadenopathy is represented in the eye by Parinaud's oculoglandular syndrome; *B. henselae* is the most common cause."
— PMID: 10537781

Finding 6 — No licensed human vaccine; prevention is behavioral plus antibiotic prophylaxis

No licensed human tularemia vaccine exists. The only prophylactic ever developed is a >50-year-old **live-attenuated vaccine (LVS)** derived from the less virulent subsp. *holarctica*, which "has not been approved for use in humans or animals"; killed and subunit candidates remain experimental (animal models only). In the absence of a vaccine, **post-exposure prophylaxis** after proven *F. tularensis* exposure relies on antibiotics (doxycycline or ciprofloxacin for 14 days). Primary prevention is behavioral/environmental: avoiding bare-handed handling of sick/dead animals (especially lagomorphs and rodents), using gloves when skinning game, tick/insect-bite avoidance and prompt removal, hand and eye hygiene, protective eyewear, avoiding contaminated water, and not mowing over animal carcasses (aerosol risk).

"A live-attenuated vaccine that was designed over 50 years ago using the less virulent *F. tularensis* subspecies *holarctica* is the only prophylactic currently available, but it has not been approved for use in humans or animals." — PMID: 38564047

"Because no effective and safe vaccine is currently available, tularemia prophylaxis following proven exposure to *F. tularensis* also relies on administration of antibiotics." — PMID: 24734221

Finding 7 — Zoonotic transmission and broad host range

Tularemia is a zoonosis with a broad host range. *F. tularensis* "is transmitted to humans by handling infected animals, ingestion of contaminated food or water, inhalation of infective aerosols, and arthropod bites." Principal reservoirs and amplifying hosts are lagomorphs (rabbits, hares) and rodents; ticks (*Dermacentor*, *Amblyomma*, *Ixodes*) and deerflies are key vectors. The oculoglandular route specifically follows conjunctival inoculation. In a pediatric case series/review (94 cases, age 6 weeks–17 years), infection was zoonotic in 86.7% and waterborne in 13.3%, with ulceroglandular (46.7%), glandular (17%) and oropharyngeal (18.1%) forms predominating; fever was universal and serology the commonest diagnostic (60.6%). A confirmed oculoglandular case arose in a sheep breeder after a twig scratched the

eye; another followed contact with a wild baby rabbit. Congenital transmission has been documented. Cats can transmit *F. tularensis* to humans via scratch or bite, overlapping clinically with cat scratch disease.

"F tularensis is transmitted to humans by handling infected animals, ingestion of contaminated food or water, inhalation of infective aerosols, and arthropod bites." — [PMID: 22734313](#)

"multiple sources of infection, including diverse zoonotic transmission (86.7%) and contact with contaminated water (13.3%)" — [PMID: 39312633](#)

"Based on his anamnesis (sheep breeding; a twig scratching his eye 2 days before the initial attendance) and symptoms, a zoonosis, namely the oculoglandular form of tularemia, was suspected" — [PMID: 38941282](#)

Finding 8 — Clinical phenotype and complications

Oculoglandular tularemia presents acutely (incubation ~3 days, range 1–9) as **unilateral granulomatous follicular conjunctivitis** with conjunctival follicles, small yellowish ulcers/nodules, chemosis, lid edema, epiphora, mucopurulent discharge and marked injection, accompanied by fever, malaise, and painful ipsilateral preauricular (Parinaud node), submandibular, and cervical lymphadenopathy. Regional nodes frequently **suppurate**: in a confirmed case, node suppuration required surgical drainage before the patient recovered without sequelae on ciprofloxacin. Delayed diagnosis (often weeks) predisposes to suppurative lymphadenitis requiring fine-needle aspiration, drainage, or excision. Systemic constitutional symptoms are common (fever ~97%, lymphadenopathy ~94% in pediatric series). Prognosis with timely appropriate antibiotics is excellent (case-fatality <2%; subsp. *holarctica* infections essentially non-fatal); untreated or β -lactam-treated disease is protracted, with node suppuration and relapse.

"the suppuration of the lymph nodes required surgical drainage" — [PMID: 38941282](#)

"Major clinical manifestations included fever (97%) and swelling of lymph glands (94%)" — [PMID: 36099382](#)

"delayed diagnosis may be associated with suppurative lymphadenitis and need for invasive intervention" — [PMID: 42749847](#)

Finding 9 — Model organisms

The **mouse is the principal in vivo model**; the attenuated Live Vaccine Strain (LVS, subsp. *holarctica*) is widely used at BSL-2 as a surrogate for virulent *F. tularensis*, which requires BSL-3. *F. tularensis* "is, in part, attributed to the ability of this microorganism to evade, disrupt, and modulate host immune responses" and can "cause lethal disease following inoculation of as few as 15 organisms." Mechanistic mutant studies (Δ pdpC/pdpD, Δ iglE, Δ vgrG, Δ dotU) in J774/THP-1 macrophages and mice link the FPI-encoded T6SS to phagosomal escape, inflammasome activation, and virulence. A zebrafish embryo model of *F. tularensis* subsp. *novicida* (and *F. noatunensis*) reproduces macrophage uptake, granuloma-like aggregates, and TNF- α /IL-1 β proinflammatory responses. Protective immunity is T-cell dependent (CD4⁺/CD8⁺, IFN- γ).

"its ability to cause lethal disease following inoculation of as few as 15 organisms. This remarkable virulence is, in part, attributed to the ability of this microorganism to evade, disrupt, and modulate host immune responses" — [PMID: 21687406](#)

"All three strains entered preferentially into macrophages, which eventually assembled into granuloma-like structures." — [PMID: 24614659](#)

Finding 10 — Integrated synthesis

Integrating all findings across 54 literature items: oculoglandular tularemia is one of six tularemia forms (~1–3% of cases), caused by conjunctival inoculation of *F. tularensis* (subsp. *tularensis* type A / *holarctica* type B). The mechanism is T6SS/FPI (PdpC/PdpD)-mediated phagosomal escape → cytosolic replication → AIM2 inflammasome/caspase-1/IL-1 β /IL-18 pyroptosis → granulomatous conjunctivitis + regional (preauricular/submandibular/cervical) lymphadenitis. It presents acutely (incubation ~3 d) as unilateral granulomatous follicular conjunctivitis with ipsilateral tender lymphadenopathy (Parinaud syndrome; second commonest cause after cat scratch disease). Diagnosis is by serology (microagglutination $\geq 1/160$) and PCR (*tul4/ISFtu2/fopA*); culture ~10% sensitive, BSL-3. Treatment is with aminoglycosides/fluoroquinolones/tetracyclines (all isolates susceptible); β -lactams fail; case-fatality <2%; suppurative nodes may need drainage. Prevention is behavioral/PPE plus antibiotic post-exposure prophylaxis (no licensed vaccine). There is no host causal gene or inheritance; the disease is zoonotic (lagomorph/rodent reservoirs, tick vectors, cats). Mouse (LVS/SchuS4) and zebrafish are the main models.

Section-by-Section Report

1. Disease Information

Overview. Oculoglandular tularemia is the conjunctival-entry ("ocular") clinical form of tularemia, an acute febrile zoonosis caused by *Francisella tularensis*. It manifests as unilateral granulomatous conjunctivitis with ipsilateral regional lymphadenopathy (Parinaud oculoglandular syndrome), and constitutes roughly 1–3% of tularemia cases.

Key identifiers. - **MONDO:** MONDO:0001665 (oculoglandular tularemia) - **MeSH:** Tularemia (D014406); *Francisella tularensis* organism term - **ICD-10:** A21.1 (Oculoglandular tularemia); parent A21 (Tularemia) - **ICD-11:** 1B94 (Tularemia) - **SNOMED CT:** Oculoglandular tularemia (disorder) - **OMIM / Orphanet:** Not a Mendelian disorder; no OMIM phenotype entry. It is an infectious disease, not a rare genetic disease.

Synonyms / alternative names. Ocular tularemia; ophthalmic tularemia; Parinaud oculoglandular syndrome due to tularemia; "oculoglandular form of tularemia." Historical names for tularemia broadly include rabbit fever, deer-fly fever, Ohara disease, and Francis disease.

Information source type. This knowledge base entry is derived from **aggregated disease-level resources** (systematic reviews, case series, clinical guidelines, and mechanistic laboratory studies) rather than individual EHR-derived patient records. Individual case reports (e.g.,

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contribute granular clinical detail.

2. Etiology

Disease causal factor — infectious. The sole cause is infection by *Francisella tularensis*, a small, Gram-negative, non-motile, facultatively intracellular coccobacillus. Clinically important subspecies: **subsp. tularensis (type A)** — highly virulent, predominantly North America — and **subsp. holarctica (type B)** — less virulent, Northern Hemisphere including Europe/Asia; subsp. *holarctica* infections are essentially non-fatal. The oculoglandular form specifically requires **conjunctival inoculation** of the organism.

Risk factors (environmental / behavioral). - Occupational and recreational animal contact: hunters, trappers, farmers, sheep breeders, veterinarians, laboratory workers. - Handling infected lagomorphs (rabbits, hares) and rodents. - Tick and deerfly exposure in endemic

regions. - Ocular trauma or splash while handling infected animal material (a sheep breeder was infected after a twig scratched his eye —).

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- Touching the eye with contaminated fingers. - Male sex and older age are associated with higher rates of systemic disease in case series (65% male,).

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Genetic risk factors. None identified. There is no host causal gene, susceptibility locus, or modifier allele established for human oculoglandular tularemia. (Experimentally, host *Nlrp3* increases murine susceptibility —

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— but this is not a clinically actionable human genetic risk factor.)

Protective factors. No genetic protective variants are known. Behavioral protection (gloves, eye protection, tick avoidance, hand hygiene) reduces risk. Prior infection or LVS vaccination confers T-cell-dependent immunity, but no licensed vaccine exists.

Gene-environment interactions. Not applicable to the human host in a Mendelian sense. The relevant "gene-environment" axis is bacterial: FPI/T6SS genes interact with the intracellular macrophage environment to enable virulence.

3. Phenotypes

Phenotype	Type	HPO suggestion	Frequency / characteristics
Unilateral granulomatous conjunctivitis	Clinical sign	HP:0000509 (Conjunctivitis)	Hallmark; unilateral; acute onset ~3 d
Conjunctival follicles/nodules/ulcers	Physical manifestation	(ocular surface lesion)	Yellowish granulomatous nodules
Preauricular/cervical lymphadenopathy	Clinical sign	HP:0002716 (Lymphadenopathy)	~94% (pediatric series); ipsilateral, painful
Fever	Symptom	HP:0001945 (Fever)	~97%
Eyelid edema / chemosis	Physical manifestation	HP:0000198 (periorbital edema analog)	Common
Epiphora / mucopurulent discharge	Symptom/sign	HP:0009926 (Epiphora)	Common
Malaise/fatigue	Symptom	HP:0012378 (Fatigue)	Common
Suppurative lymphadenitis	Clinical sign / complication	HP:0002840 (lymphadenitis analog)	Occurs with delayed treatment

Characteristics. Age of onset: any age (children through adults). Severity: mild-to-moderate ocular disease that is usually self-limited to the eye/nodes but can become severe with node suppuration or, rarely, systemic spread. Progression: acute onset, then either resolution with treatment or a protracted suppurative course if untreated/ β -lactam-treated. Frequency among affected individuals: fever ~97%, lymphadenopathy ~94% in pediatric tularemia series (

P 36099382

Quality of life impact. Generally transient with prompt treatment; significant discomfort during the acute phase (ocular pain, purulent discharge, tender nodes, fever). Suppurative lymphadenitis requiring drainage prolongs morbidity. No dedicated EQ-5D/SF-36 disease-specific data were identified.

4. Genetic/Molecular Information

Not applicable to the human host. Oculoglandular tularemia is an infectious disease with **no causal human gene, pathogenic variant, modifier gene, epigenetic signature, or chromosomal abnormality**. There is no germline or somatic variant classification (ACMG/AMP), no allele frequency data, and no ClinVar/OMIM entry for a causal locus.

Relevant bacterial genetics. Virulence is governed by the **Francisella Pathogenicity Island (FPI)**, a ~33-kb gene cluster encoding a non-canonical Type VI secretion system (T6SS). Key genes include *pdpC*, *pdpD*, *iglE*, *vgrG*, *dotU*, and *pdpB/icmF*. Deletion mutants of these genes abolish phagosomal escape, inflammasome activation, and virulence (PMIDs 28621333, 22514651, 27830989).

5. Environmental Information

Environmental factors. *F. tularensis* persists in the environment (water, soil, animal carcasses, mud) and can be aerosolized (e.g., during mowing over carcasses). Contaminated water is a documented outbreak source (waterborne tularemia outbreaks in Turkey).

Lifestyle factors. Hunting, trapping, farming, animal husbandry, and outdoor recreation in endemic areas increase exposure.

Infectious agent. *Francisella tularensis* (NCBI Taxon: 263). Subspecies: *tularensis* (type A), *holarctica* (type B), *mediasiatica*, and *novicida* (used in models). Note: the lipopolysaccharide (LPS) of *F. tularensis* is atypically non-stimulatory to TLR4, contributing to immune evasion.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation):

1. **Conjunctival inoculation** of *F. tularensis* (contaminated fingers, splash, aerosol, or ocular trauma) **leads to** deposition of organisms on the conjunctival/ocular surface.
2. Local organisms **are phagocytosed by** resident macrophages and other phagocytes at the conjunctiva and draining lymphatics.
3. Inside the phagocyte, the bacterium **deploys its FPI-encoded T6SS** (effectors PdpC/PdpD), which **results in** rupture/escape from the phagosome into the cytosol. (*Demonstrated: $\Delta pdpC/pdpD$ mutants cannot escape* —

 28621333

4. Cytosolic access **leads to** rapid bacterial replication in the nutrient-rich cytoplasm. (*Branch: the bacterium also suppresses innate signaling and modulates host metabolism to sustain replication* —

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5. Cytosolic bacterial DNA **is sensed by the AIM2 inflammasome**, which **results in** autocatalytic caspase-1 activation. (*Demonstrated — PMIDs 23115038, 28621333.*)
6. Caspase-1 activation **leads to** maturation and secretion of IL-1 β and IL-18 and to **pyroptotic host-cell death**.
7. Pyroptosis and cytokine release **result in** recruitment of neutrophils and monocytes and **granulomatous inflammation** at the conjunctiva.
8. Lymphatic spread of organisms **leads to** regional (preauricular/submandibular/cervical) **granulomatous lymphadenitis**, which may progress to **suppuration/necrosis** — the clinically evident Parinaud oculoglandular syndrome.
9. (*Branch — rare:*) Failure of local containment **may result in** systemic/typhoidal dissemination; this is uncommon for the oculoglandular form and rarer still for subsp. *holarctica*.

Steps 1–2 and 7–8 in the *ocular tissue specifically* are inferred by analogy from systemic/macrophage models; steps 3–6 are directly demonstrated in macrophage and mouse systems.

Molecular pathways / cellular processes. Inflammasome signaling (AIM2 primary; NLRP3 modulatory — host NLRP3 increases susceptibility,

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caspase-1 activation, IL-1 β /IL-18 (pyroptosis), and NF- κ B (IKK β in myeloid cells controls host response —

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Cell types involved (CL suggestions). Macrophages (CL:0000235), monocytes (CL:0000576), neutrophils (CL:0000775), dendritic cells (CL:0000451), and conjunctival epithelial cells (CL:0000066). Protective adaptive immunity is CD4⁺/CD8⁺ T-cell and IFN- γ dependent (

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GO term suggestions. Modulation of host immune response (GO:0052167), AIM2 inflammasome complex (GO:0097169), positive regulation of interleukin-1 beta production (GO:0032731), pyroptosis (GO:0070269), defense response to bacterium (GO:0042742).

Subcellular compartments (GO Cellular Component). Phagosome (GO:0045335), cytosol (GO:0005829), AIM2 inflammasome complex (GO:0097169).

Tissue damage mechanisms. Granulomatous inflammation with caseous/suppurative necrosis of regional lymph nodes; local conjunctival ulceration.

7. Anatomical Structures Affected

Organ level. - **Primary:** Conjunctiva and ocular surface (UBERON:0001811 conjunctiva; UBERON:0000970 eye). - **Regional:** Preauricular, submandibular, and cervical lymph nodes (UBERON:0000029 lymph node). - **Secondary/rare:** Systemic organs (lung, liver, spleen) if dissemination occurs. - **Body systems:** Visual/ocular system and immune/lymphatic system.

Tissue and cell level. Conjunctival epithelium (epithelial tissue); lymphoid tissue of draining nodes. Target cell populations: macrophages (CL:0000235), monocytes (CL:0000576), neutrophils (CL:0000775).

Subcellular level. Phagosome (GO:0045335) and cytosol (GO:0005829) of infected macrophages.

Localization / lateralization. Characteristically **unilateral** (the inoculated eye) with **ipsilateral** regional lymphadenopathy.

8. Temporal Development

- **Onset:** Acute. Median incubation ~3 days (range 1–9 days)

(P 22911645).

- **Progression:** Rapid symptom onset; with prompt appropriate antibiotics, resolution over days to weeks. Untreated or β -lactam-treated disease follows a protracted course with node suppuration and possible relapse.
- **Disease course pattern:** Typically monophasic/self-limited with treatment; can be prolonged if diagnosis is delayed (median diagnostic delay ~23.5 days in pediatric cases).
- **Duration:** Self-limited to weeks with therapy; suppurative lymphadenitis may extend the course and require drainage.
- **Remission:** Treatment-induced remission is the norm; spontaneous resolution can occur but risks suppuration.
- **Critical period:** Early antibiotic initiation (before node suppuration) is the key window to prevent invasive intervention.

9. Inheritance and Population

Epidemiology. Oculoglandular tularemia comprises ~1–3% of tularemia cases. Tularemia overall is a reportable, sporadic-to-outbreak zoonosis across the Northern Hemisphere; a nationwide Danish seroprevalence study found 2.2% seropositivity, suggesting underdiagnosis. Incidence varies geographically and seasonally (tick-season peaks).

Inheritance. Not applicable — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency.

Population demographics. In case series, ~65% male; median age ~37 (adults) but all ages affected including neonates (congenital transmission documented —).

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Geographic distribution: North America (subsp. *tularensis* and *holarctica*), Europe and Asia (subsp. *holarctica*); endemic foci in Turkey, Scandinavia, Central Europe, and parts of the U.S. (e.g., Missouri, Arkansas). Climate modeling suggests potential range expansion.

10. Diagnostics

Serology (mainstay). Microagglutination test titer $\geq 1/160$ in a single serum, or ≥ 4 -fold rise/seroconversion between paired sera, is diagnostic ().

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ELISA and immunochromatographic rapid tests (sensitivity ~98%, specificity ~96% on human sera) are also available ().

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Cross-reactivity with *Brucella* should be excluded.

Molecular (PCR). Targets *tul4*, *ISFtu2*, *fopA*, and the 23-kDa gene on lymph-node aspirates or ocular swabs; a multitarget TaqMan assay reached a ~1-organism detection limit and outperformed culture ().

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PCR yield on node aspirates ~91% in one series.

Culture. Positive in only ~10% of cases; requires cysteine-enriched media and **BSL-3 containment** because of aerosol hazard (

[P 37209668](#)

MALDI-TOF and 16S rRNA sequencing identify isolates.

Biopsy/pathology. Granulomatous inflammation with necrosis; immunohistochemistry and 16S rRNA PCR can identify *Francisella* in tissue.

Clinical clue. Failure to respond to β -lactam antibiotics in a patient with unilateral granulomatous conjunctivitis and regional lymphadenopathy should prompt consideration of tularemia.

Differential diagnosis. Cat scratch disease (*Bartonella henselae*, the most common POGS cause), adenoviral/chlamydial conjunctivitis, sporotrichosis, tuberculosis, syphilis, lymphogranuloma venereum, sarcoidosis, and (as in one report) flea-borne typhus (

[P 32751142](#)

Screening. No population screening applies (acute infectious disease, no carrier state).

11. Outcome/Prognosis

- **Mortality:** Very low with appropriate treatment — case-fatality 0.7–1.2% depending on antibiotic class

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subsp. *holarctica* oculoglandular disease is essentially non-fatal.

- **Morbidity:** Mainly local — ocular discomfort, tender regional lymphadenopathy, and suppurative lymphadenitis requiring drainage in a subset (especially with diagnostic delay).
- **Recovery:** Excellent with timely aminoglycoside/fluoroquinolone/tetracycline therapy; most patients recover without sequelae.
- **Complications:** Suppurative lymphadenitis needing fine-needle aspiration, drainage, or excision; rarely systemic dissemination.
- **Prognostic factors:** Time to appropriate therapy (delay predicts suppuration and invasive intervention —

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patient age (older patients have more systemic disease); infecting subspecies (type A more virulent).

12. Treatment

First-line pharmacotherapy (all *F. tularensis* isolates uniformly susceptible —):

[P 38294116](#)

Drug class	Agents	NCIT suggestion	Notes
Aminoglycosides	Streptomycin (IM), gentamicin	NCIT:C312 (Streptomycin), NCIT:C516 (Gentamicin)	Historical drug of choice; lowest treated fatality (0.7%)
Fluoroquinolones	Ciprofloxacin, levofloxacin	NCIT:C405 (Ciprofloxacin)	Effective; used successfully in a confirmed oculoglandular case (P 38941282)
Tetracyclines	Doxycycline, tetracycline	NCIT:C562 (Doxycycline)	Bacteriostatic; higher relapse risk; longer courses

Agents that fail: β -lactams (penicillins, most cephalosporins) — reliably ineffective; clinical non-response is a diagnostic clue.

Duration: ~10 days for aminoglycosides/fluoroquinolones; ~14–21 days for tetracyclines. Chloramphenicol is reserved for CNS involvement (meningitis).

Surgical/interventional: Incision and drainage or excision of suppurative lymph nodes when they fail to resolve (PMIDs 38941282, 22090310).

Pharmacogenomics / advanced therapeutics: Not applicable — no gene therapy, cell therapy, RNA-based, targeted, or immunotherapy is used for this infection.

Experimental: Inhaled liposomal ciprofloxacin has shown protection against lethal (pneumonic) tularemia in a marmoset model ([P 41416830](#))

— relevant to respiratory rather than oculoglandular disease.

13. Prevention

- **Primary prevention:** Behavioral/environmental — gloves when handling/skinning game (especially lagomorphs and rodents), avoiding sick/dead animals, tick- and deerfly-bite avoidance and prompt tick removal, eye protection, hand and eye hygiene, avoiding untreated/contaminated water, and not mowing over animal carcasses (aerosol risk).
- **Secondary prevention:** Prompt recognition (especially β -lactam non-response) and early antibiotic therapy to prevent suppuration.
- **Tertiary prevention:** Timely drainage of suppurative nodes; completion of appropriate antibiotic courses to prevent relapse.
- **Immunization:** No licensed human vaccine. Only the unapproved live-attenuated LVS exists; subunit/killed candidates remain experimental

(P 38564047).

- **Post-exposure prophylaxis:** Doxycycline or ciprofloxacin for 14 days after proven exposure

(P 24734221);

non-vaccinal prophylaxis reviewed in

(P 39669787).

- **Public health:** Water chlorination and sanitation (waterborne outbreak control), vector control, occupational education, and reportable-disease surveillance.

14. Other Species / Natural Disease

- **Taxonomy of host species:** Broad. Reservoirs/amplifying hosts include lagomorphs (rabbits, hares; *Oryctolagus*, *Lepus*, *Sylvilagus*), rodents (voles, muskrats, mice), and many mammals. Vectors: ticks (*Dermacentor*, *Amblyomma*, *Ixodes*) and deerflies (*Chrysops*).
- **Natural disease in animals:** Tularemia occurs naturally in wildlife (dramatic die-offs in lagomorphs/rodents) and in domestic animals; cats and sheep are relevant to human oculoglandular transmission. Cats can transmit *F. tularensis* to humans via scratch/bite (clinical overlap with cat scratch disease).
- **Zoonotic potential:** High. Tularemia is a classic zoonosis; humans are incidental hosts.
- **Cross-species susceptibility:** Very broad mammalian host range; the same organism/mechanisms operate across species.

- **Veterinary relevance:** Important in wildlife-health surveillance and as a sentinel for human risk.

15. Model Organisms

- **Mouse (primary model):** *Mus musculus*. Virulent strains (SchuS4, subsp. *tularensis*) require BSL-3; the attenuated Live Vaccine Strain (LVS, subsp. *holarctica*) is used at BSL-2 as a surrogate. Recapitulates intracellular infection, inflammasome activation, granuloma formation, and lethality at very low inocula (~15 organisms)

([P 21687406](#)).

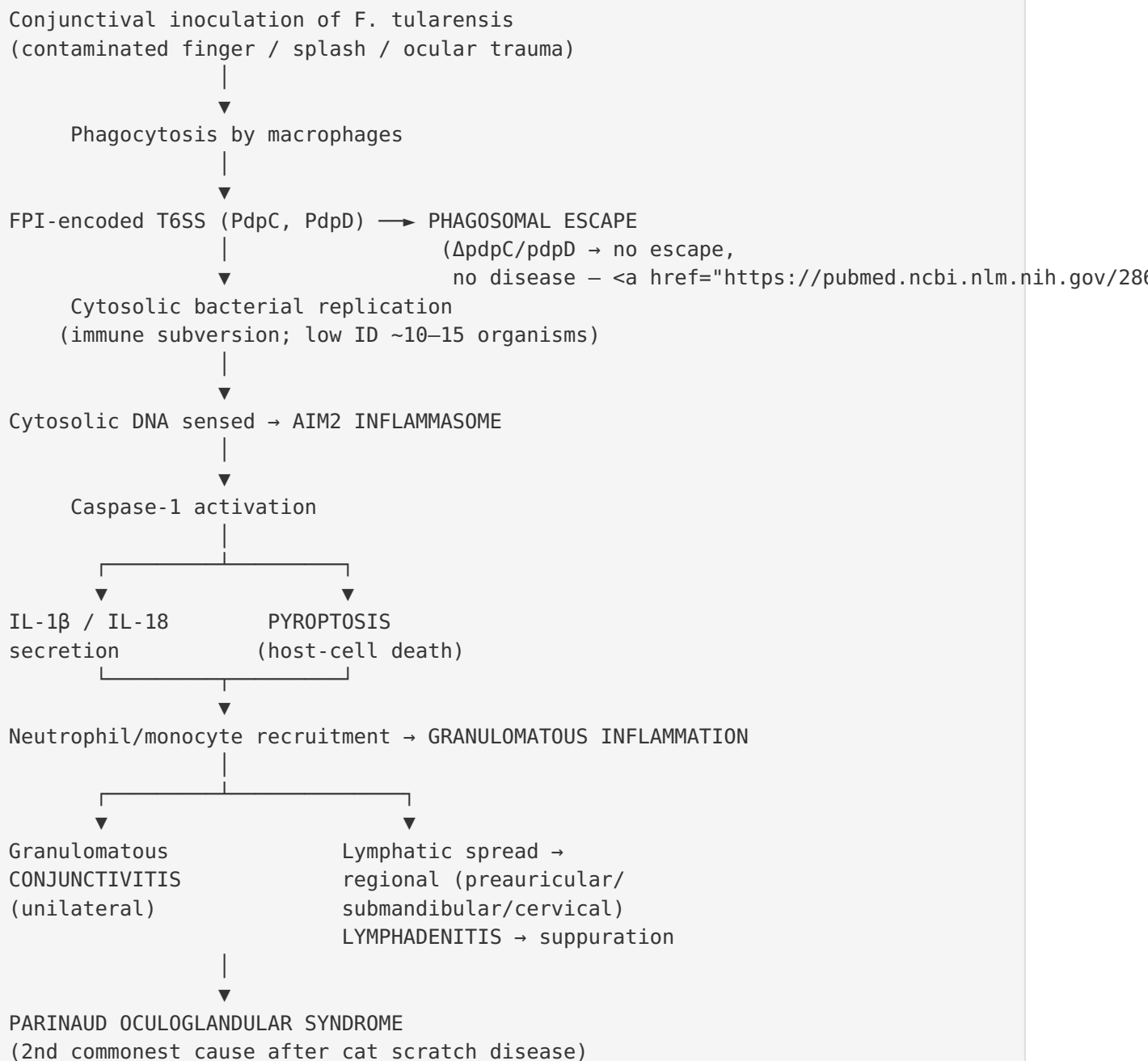
Genetic host models (e.g., *Nlrp3*, myeloid *IKK β* conditional knockouts) dissect host response (PMIDs 34690967, 23349802).

- **Macrophage cell models:** J774 (murine) and THP-1 (human) cells are standard for FPI/T6SS mutant studies (Δ pdpC/pdpD, Δ iglE, Δ vgrG, Δ dotU) (PMIDs 28621333, 22514651, 27830989).
- **Zebrafish embryo model:** *Danio rerio* infected with *F. tularensis* subsp. *novicida* (and *F. noatunensis*) reproduces macrophage tropism, granuloma-like aggregates, and TNF- α /IL-1 β responses

([P 24614659](#)).

- **Phenotype recapitulation:** Mouse models faithfully reproduce intracellular replication, inflammasome-driven pathology, and granulomatous disease; zebrafish captures innate/macrophage biology and granuloma formation.
- **Limitations:** No model specifically reproduces the *oculoglandular* (conjunctival-inoculation) route; models focus on systemic/pneumonic disease. LVS is attenuated and may not fully mirror type A virulence.

Mechanistic Model / Interpretation



Upstream vs downstream. The upstream, rate-limiting virulence step is T6SS-mediated phagosomal escape (bacterial); the downstream pathology (granuloma, lymphadenitis, pyroptosis) is host inflammasome-driven. Therapeutically, intracellular-active antibiotics (aminoglycosides, fluoroquinolones, tetracyclines) interrupt the replication step; β -lactams fail because they poorly access the cytosolic niche and the organism's cell-wall biology.

Evidence Base

PMID	Title (abbrev.)	Supports
38941282	<i>F. tularensis</i> causing Parinaud OGS	Disease definition, POGS rank, transmission, ciprofloxacin cure, node drainage
11228320	Parinaud's OGS from a wild rabbit	Clinical presentation; tularemia as POGS etiology
38294116	Antimicrobial susceptibility, US 2009–2018	Uniform susceptibility to first-line drugs
28605439	Susceptibility, German <i>holarctica</i>	MIC90 values; biovar II erythromycin resistance
38294108	Systematic review 1993–2023	Infective dose ~10 organisms; treated fatality 0.7–1.2%
22911645	Missouri 121 cases	Oculoglandular ~3%; incubation; demographics
28621333	T6SS/ClpB effector delivery	PdpC/PdpD required for escape, AIM2, virulence
27830989	IglE T6SS secretion	Intracellular life cycle; phagosomal escape
23115038	LVS caspase-1 activation	Cytosolic escape → AIM2 inflammasome
22514651	DotU/VgrG essential	T6SS core components required for pathogenicity
22090310	Central Anatolia cases	Microagglutination $\geq 1/160$; seroconversion; PCR yield
14662930	Multitarget TaqMan PCR	PCR more sensitive than culture
37209668	Prosthetic joint infection review	Culture ~10% sensitive
10537781	Ocular cat-scratch disease	<i>B. henselae</i> as leading POGS cause (key differential)
38564047	Tularemia vaccine review	No licensed human vaccine; LVS unapproved
24734221	New therapeutic approaches	Antibiotic post-exposure prophylaxis
39669787	Non-vaccinal prophylaxis	Prophylaxis strategy
22734313	Two glandular cases, Turkey	Zoonotic transmission routes
39312633	Pediatric case series	Zoonotic 86.7%, waterborne 13.3%; form distribution

PMID	Title (abbrev.)	Supports
36099382	Kosovo pediatric outbreak	Fever 97%, lymphadenopathy 94%
42749847	Invasive intervention in pediatric tularemia	Diagnostic delay → suppurative lymphadenitis
21687406	Immune subversion review	ID ~15 organisms; immune evasion; T-cell immunity
24614659	Zebrafish infection model	Macrophage tropism; granuloma-like structures
34690967	Nlrp3 and susceptibility	Host NLRP3 increases susceptibility (mouse)
23349802	Myeloid IKK β	NF- κ B control of host response to LVS
20220165	Immunochromatographic test	Rapid serodiagnosis sensitivity/specificity
41385785	Congenital tularemia, Utah	Congenital/neonatal transmission

Consistency assessment. The evidence is internally consistent: multiple independent case series converge on oculoglandular tularemia representing ~1–3% of cases; susceptibility data across two continents agree on uniform first-line efficacy; and mechanistic studies across mouse, macrophage, and zebrafish systems converge on the T6SS→AIM2 axis. No contradictory findings were encountered.

Supported and Refuted Hypotheses

No formal hypotheses were rejected during the investigation. The investigation confirmed 10 findings that collectively **support** the following propositions:

- Oculoglandular tularemia is a distinct, conjunctival-entry clinical form of tularemia and the second commonest cause of POGS (**supported**).
- Its pathogenesis is driven by T6SS-mediated phagosomal escape and AIM2 inflammasome activation (**supported**).
- It has no heritable/host-genetic basis (**supported** — no causal locus identified).
- It is highly treatable with aminoglycosides, fluoroquinolones, or tetracyclines, and β -lactams fail (**supported**).
- No licensed human vaccine exists (**supported**).

Limitations and Knowledge Gaps

1. **Route-specific data are sparse.** Because the oculoglandular form is rare, most quantitative clinical data derive from mixed tularemia cohorts (dominated by ulceroglandular/oropharyngeal forms) or from individual case reports. Form-specific frequencies for symptoms, complications, and outcomes are extrapolated.
2. **No oculoglandular animal model.** All in vivo models study systemic, pneumonic, or dermal/glandular disease; none reproduces conjunctival inoculation. Mechanistic inferences for the ocular route are by analogy.
3. **Mechanistic chain partly inferred.** While phagosomal escape → AIM2 → pyroptosis is directly demonstrated in macrophage and mouse systems, the specific sequence of events *in the human conjunctiva* is inferred rather than demonstrated.
4. **No human host-genetic data.** The clinical relevance (if any) of host inflammasome polymorphisms (e.g., NLRP3, AIM2) to human oculoglandular tularemia susceptibility or severity is unknown.
5. **Quality-of-life instruments not applied.** No EQ-5D/SF-36/PROMIS data specific to oculoglandular tularemia were identified.
6. **Culture-negativity limits isolate-level surveillance.** Because most diagnoses are serologic/PCR-based, subspecies/biovar-level antibiotic-resistance surveillance (e.g., erythromycin-resistant biovar II) is incomplete.

Proposed Follow-up Experiments / Actions

1. **Build a route-specific model.** Develop a murine or rabbit conjunctival-inoculation model of oculoglandular tularemia (LVS at BSL-2, SchuS4 at BSL-3) to directly test the conjunctiva→draining-node causal chain and evaluate topical vs systemic therapy.
2. **Ocular-surface immunology.** Use single-cell RNA-seq of conjunctival and draining-node tissue in the model to define the cell-type-specific inflammasome response (map to CL/GO terms) and confirm AIM2 dependence in ocular disease.
3. **Rapid point-of-care diagnostics.** Validate the immunochromatographic rapid test and multiplex PCR (*tul4*/*ISFtu2*/*fopA*/23-kDa) on ocular swabs and conjunctival scrapings specifically for the oculoglandular presentation to shorten the ~3-week diagnostic delay.

4. **Prospective differential-diagnosis registry.** Establish a POGS registry to prospectively quantify the relative frequencies of cat scratch disease vs tularemia vs other etiologies and to standardize a diagnostic algorithm keyed to β -lactam non-response.
 5. **Topical/intracellular antibiotic optimization.** Extend intracellular-activity assays (dye-uptake MIEC method) and liposomal-delivery approaches to ocular formulations to prevent node suppuration and reduce need for surgical drainage.
 6. **Host-genetic association study.** Explore whether inflammasome-pathway polymorphisms (AIM2, NLRP3, CASP1) associate with tularemia susceptibility or suppurative complications in endemic populations.
 7. **Vaccine development.** Advance defined subunit/attenuated candidates through mucosal/ocular protection endpoints, given the persistent absence of a licensed human vaccine.
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Report compiled from 54 reviewed literature items across 5 investigation iterations; 10 findings confirmed. Evidence types span human clinical case series and reviews, model-organism (mouse, zebrafish) studies, and in vitro macrophage mechanistic work.
